



Chapter 5

Analog Transmission

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5-1 DIGITAL-TO-ANALOG CONVERSION

***Digital-to-analog** conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data.*

Topics discussed in this section:

Aspects of Digital-to-Analog Conversion

Amplitude Shift Keying

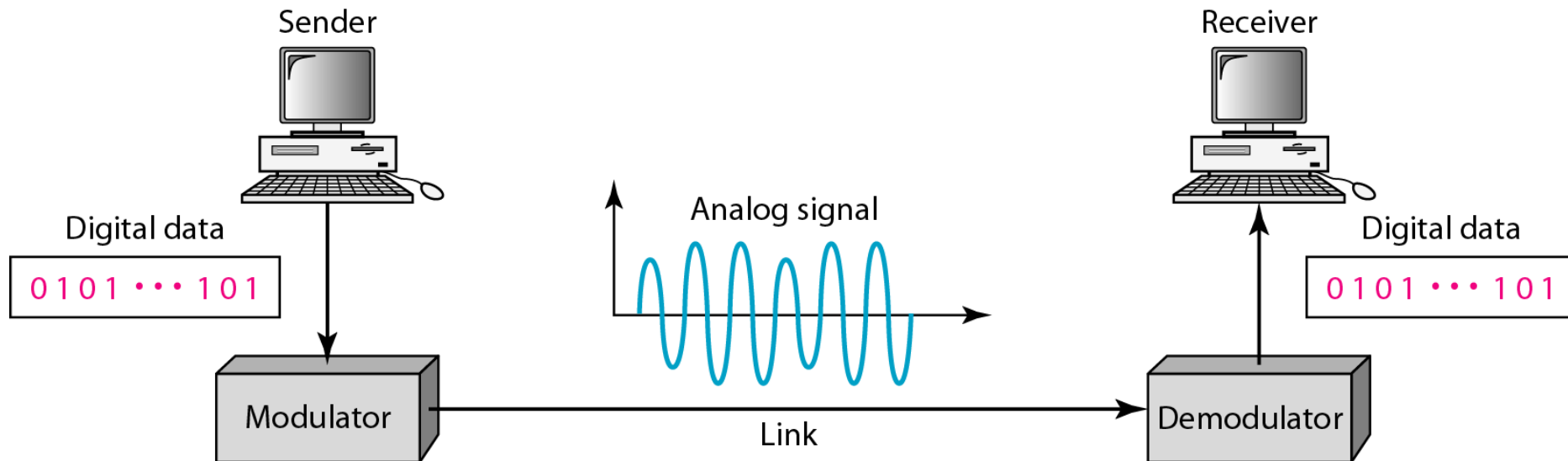
Frequency Shift Keying

Phase Shift Keying

Quadrature Amplitude Modulation

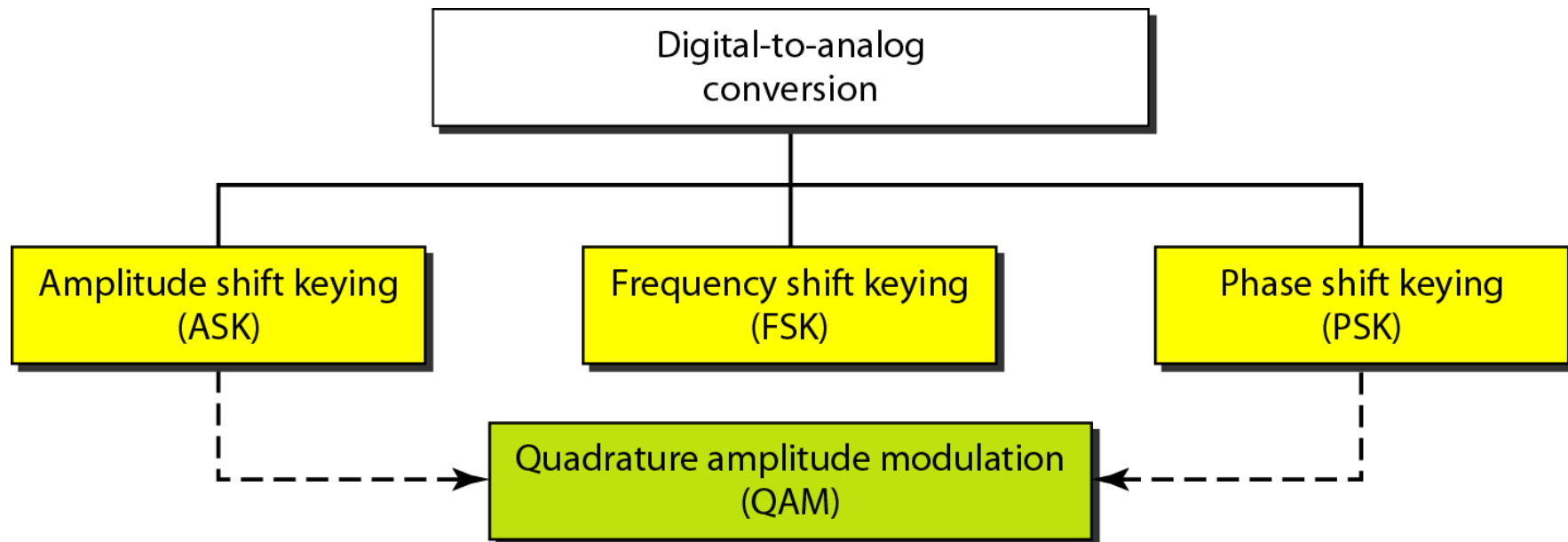
Digital-to-analog conversion

- Digital-to-analog conversion is the process of changing one of the characteristics of an analog signal based on the information in digital data.



Types of Digital-to-analog conversion

- A sine wave is defined by three characteristics: amplitude, frequency, and phase.
- Among any of the three characteristics can be used to for modulating digital data into an analog signal: amplitude shift keying (ASK), frequency shift keying (FSK), and phase shift keying (PSK).



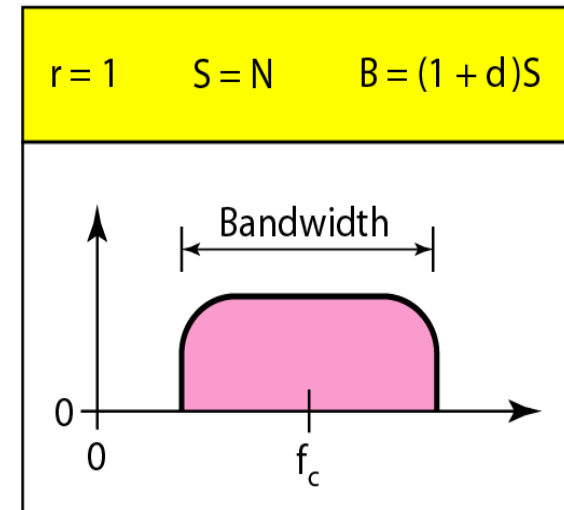
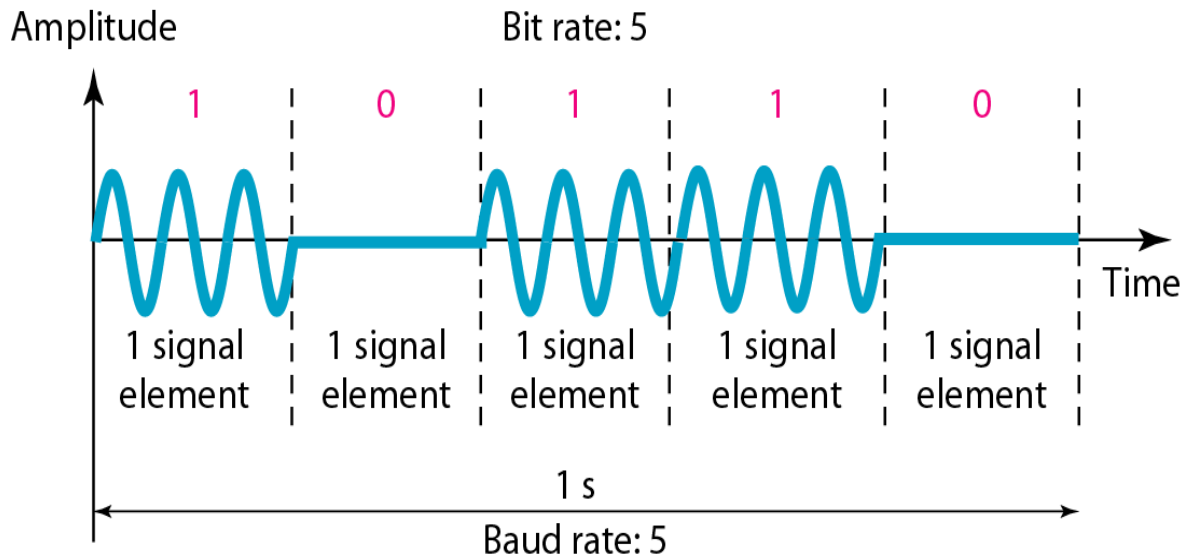
Note

**Bit rate is the number of bits per second.
Baud rate is the number of signal
elements per second.**

**In the analog transmission of digital
data, the baud rate is less than
or equal to the bit rate.**

Binary amplitude shift keying(BASK)

- In amplitude shift keying(ASK), the amplitude of the carrier signal is varied to create signal elements. Both frequency and phase remain constant while the amplitude changes.
- Binary ASK is normally implemented using only two levels, that's why it is referred to as referred to as binary amplitude shift keying or on-off keying (OOK).



Bandwidth for BASK

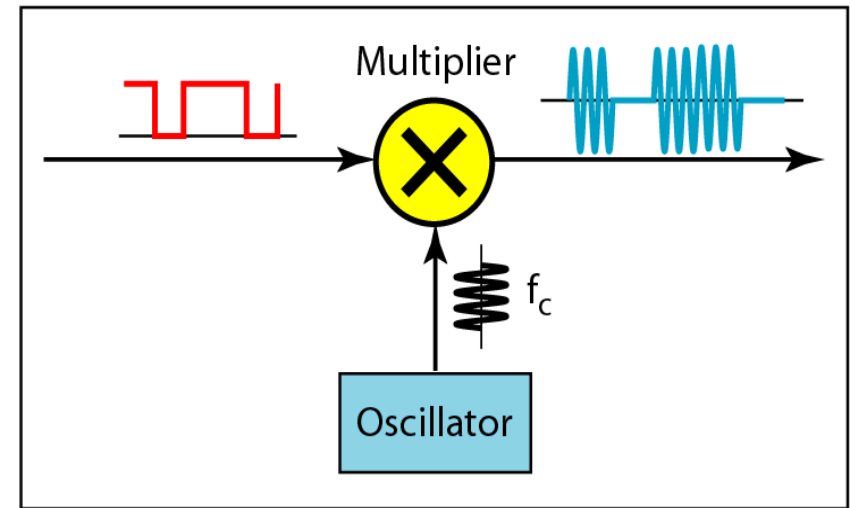
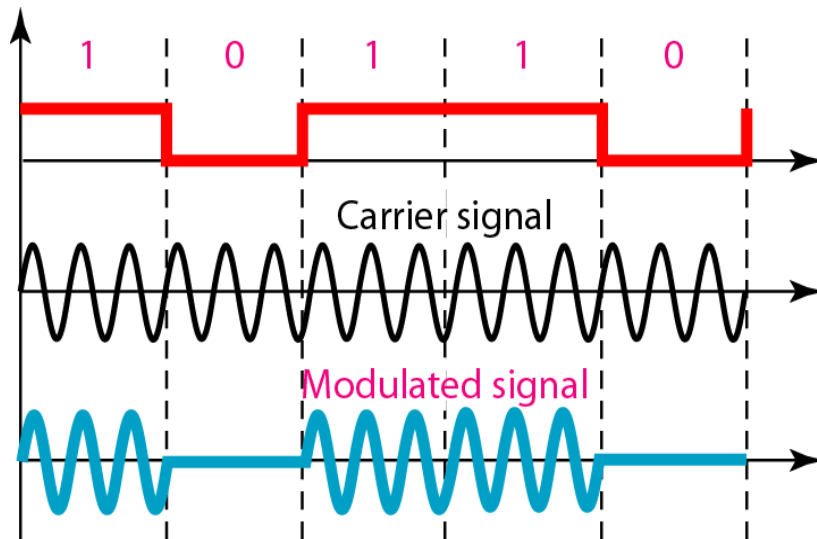
- Bandwidth is proportional to signal rate(Baud rate).
- Modulation index factor called d , which depends on the modulation and filtering process.
- The value of d is between 0 and 1.
- This bandwidth B can be expressed as shown, where S is the signal rate of the signal.

$$B = (1 + d) \times S$$

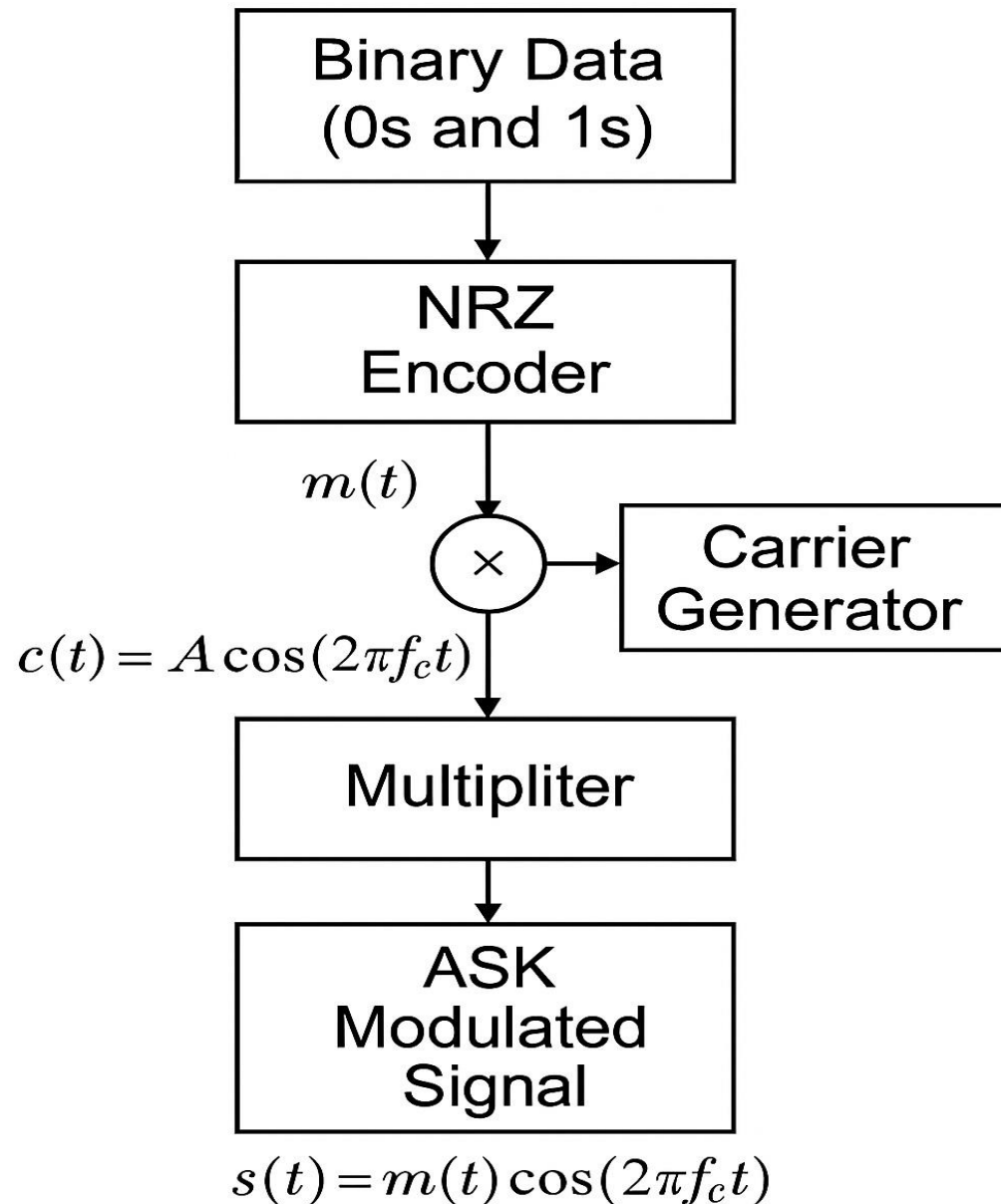
- The formula shows that the required bandwidth has a minimum value of S and a maximum value of $2S$.

Implementation of binary ASK

- If digital data are presented as a unipolar NRZ digital signal with a high voltage of 1 V and a low voltage of 0 V.
- The implementation can be achieved by multiplying the NRZ digital signal by the carrier signal coming from an oscillator.
- When the amplitude of the NRZ signal is 1, the amplitude of the carrier frequency is held, and
- When the amplitude of the NRZ signal is 0, the amplitude of the carrier frequency is zero.



Implementation of binary ASK

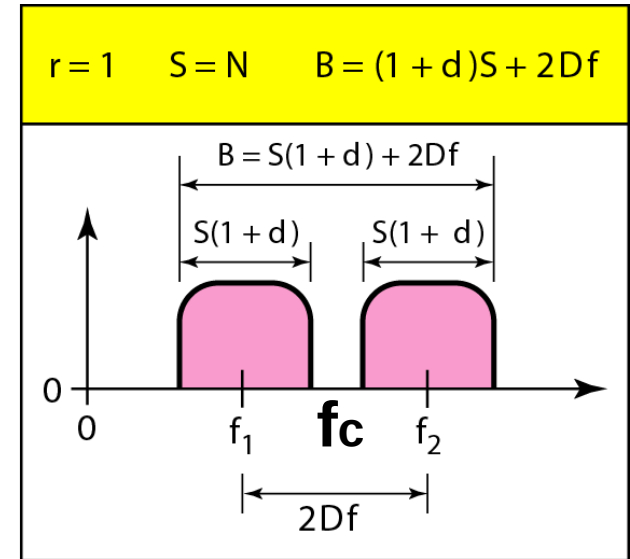
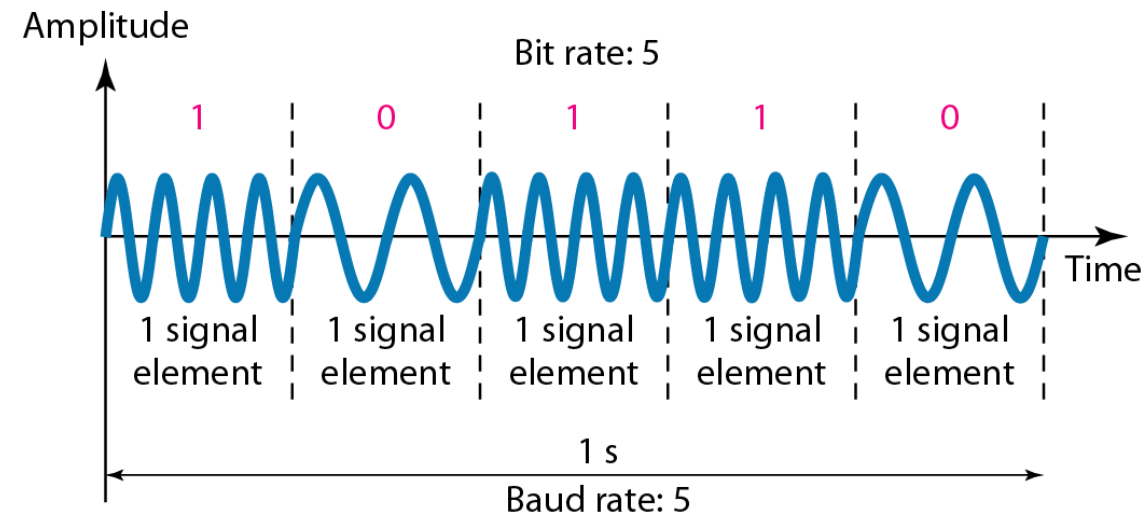


Binary frequency shift keying(BFSK)

- In frequency shift keying, the frequency of the carrier signal is varied to represent data.
- The frequency of the modulated signal is constant for the duration of one signal element, but changes for the next signal element if the data element changes.
- Both peak amplitude and phase remain constant for all signal elements.
- In binary FSK (or BFSK), considered two carrier frequencies f_1 and f_2 .
- We can think of FSK as two ASK signals, each with its own carrier frequency (f_1 or f_2).
- The first carrier f_1 is used when the data element is 0; and the second f_2 used for the data element 1.
- If the difference between the two frequencies is $2\Delta f$, then the required bandwidth is

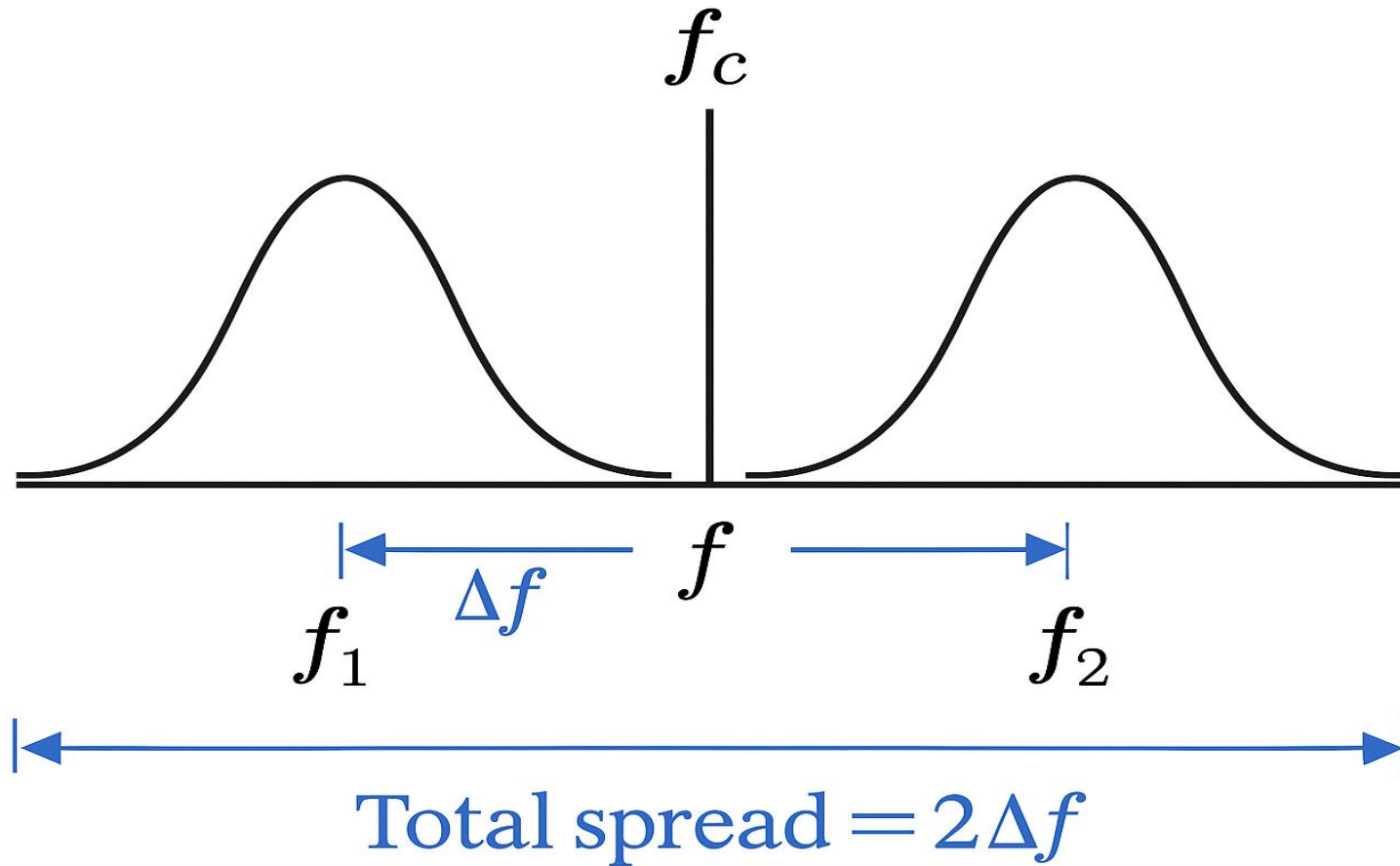
$$B = (1 + d) \times S + 2\Delta f$$

Binary frequency shift keying(BFSK)



- The first carrier f_1 is used if the data element is 0; and f_2 used if the data element is 1.
- The middle of one bandwidth is f_1 and the middle of the other is f_2 .
- Both f_1 and f_2 are Δf apart from the midpoint between the two bands means $f_1 = f_c - \Delta f$ and $f_2 = f_c + \Delta f$.
- The difference between the two frequencies, $f_2 - f_1 = f_c + \Delta f - f_c + \Delta f = 2\Delta f$.

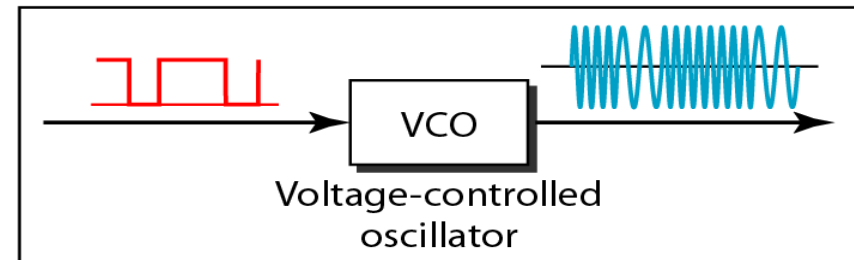
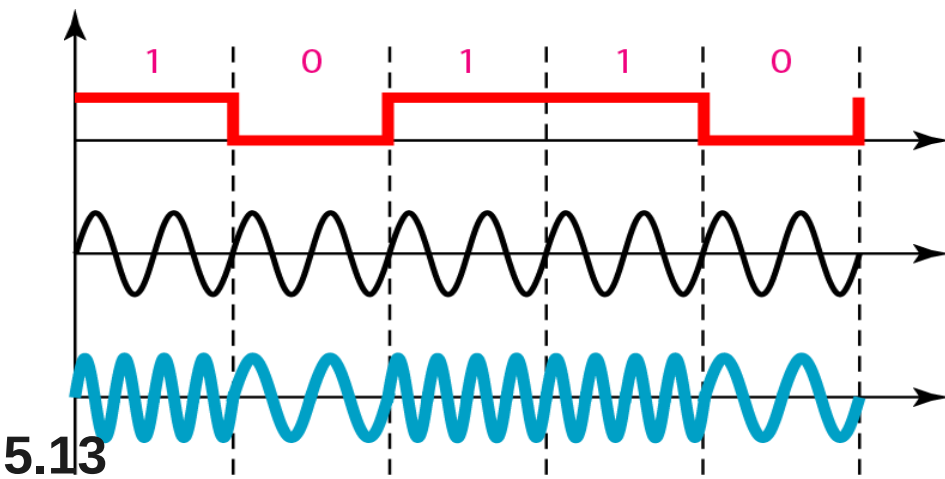
Binary frequency shift keying(BFSK)



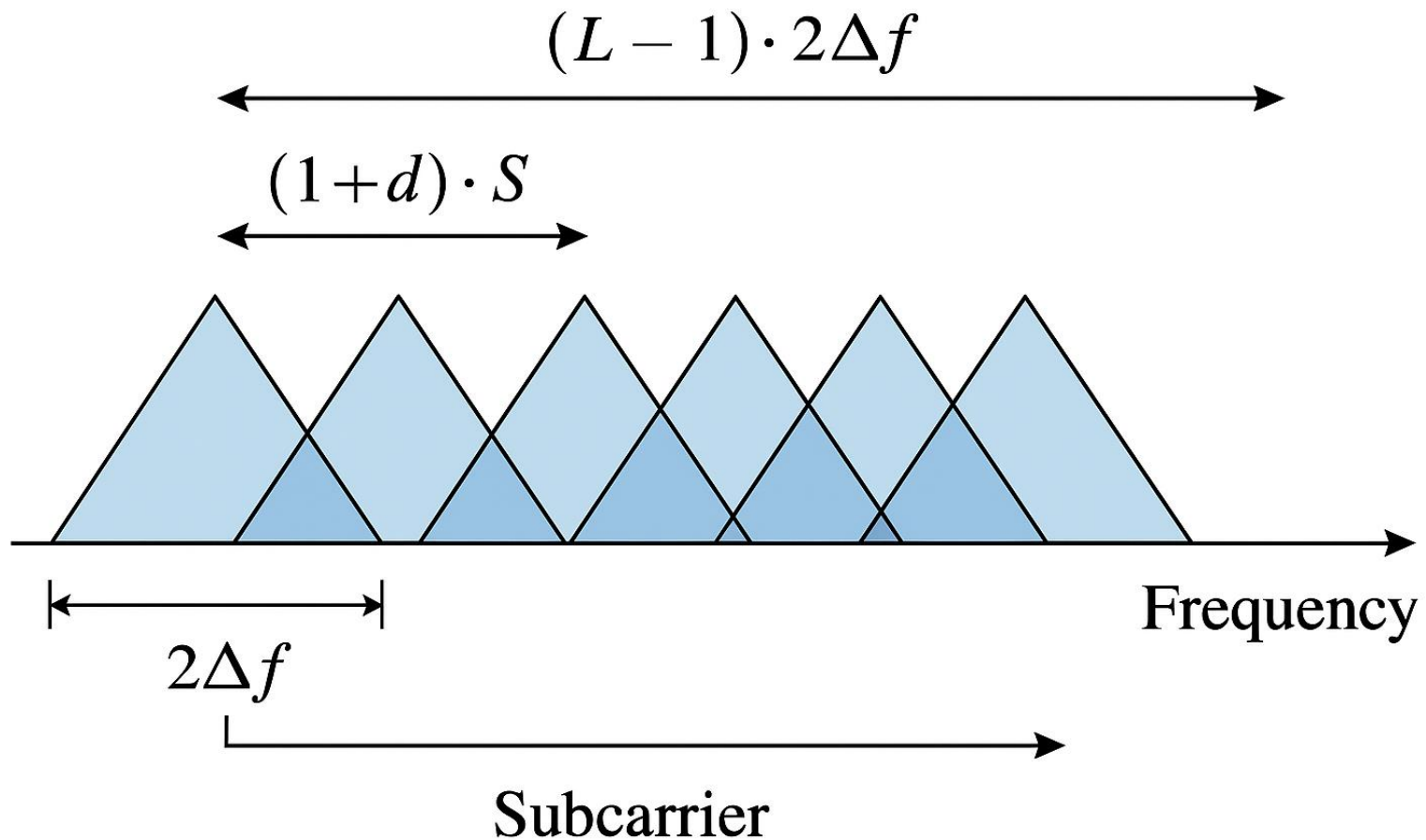
The difference between the two frequencies, $f_2 - f_1 = f_c + \Delta f - f_c + \Delta f = 2\Delta f$

Multilevel frequency shift keying(MFSK)

- Coherent BFSK can be implemented by one voltage-controlled oscillator (VCO) that changes its frequency according to the input voltage.
- The input to the oscillator is the unipolar NRZ signal.
- When the amplitude of NRZ is zero, the oscillator **keeps its regular frequency**, and
- When the amplitude is positive, the frequency **is increased**.
- We can use more than two frequencies. For example, we can use four different frequencies f_1 , f_2 , f_3 , and f_4 to send 2 bits at a time, to send 3 bits it needs 8 frequencies and so on.
- The frequencies need to be $2\Delta f$ apart.



Multilevel frequency shift keying(MFSK)



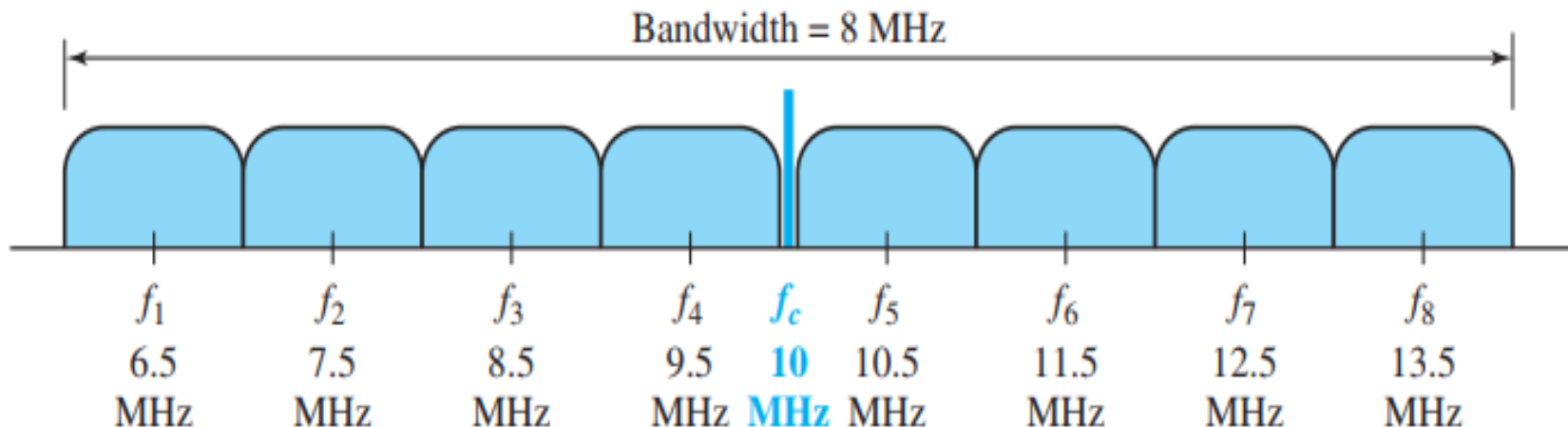
Multilevel frequency shift keying(MFSK)

We need to send data 3 bits at a time at a bit rate of 3 Mbps. The carrier frequency is 10 MHz. Calculate the number of levels (different frequencies), the baud rate, and the bandwidth.

Solution

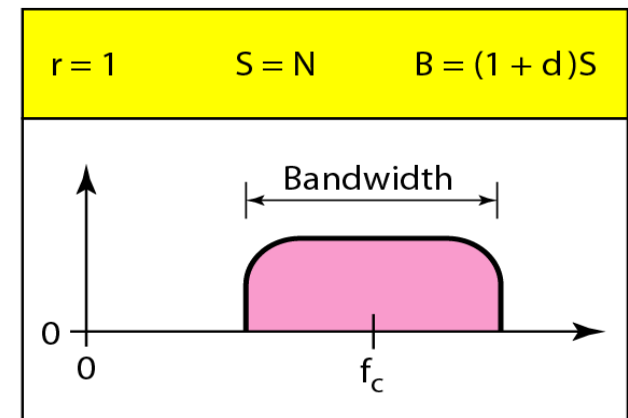
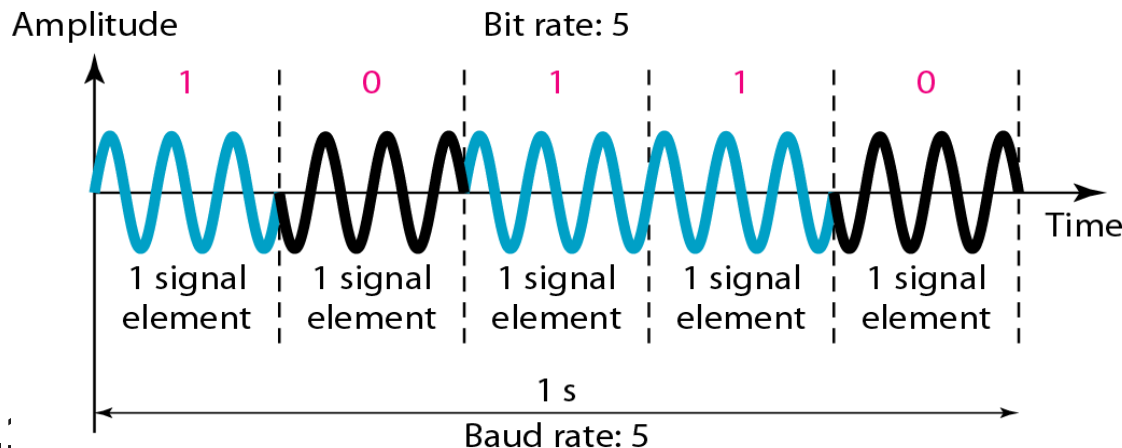
We can have $L = 2^3 = 8$. The baud rate is $S = 3 \text{ MHz}/3 = 1 \text{ Mbaud}$. This means that the carrier frequencies must be 1 MHz apart ($2\Delta_f = 1 \text{ MHz}$). The bandwidth is $B = 8 \times 1 = 8 \text{ MHz}$. Figure 5.8 shows the allocation of frequencies and bandwidth.

Figure 5.8 *Bandwidth of MFSK used in Example 5.6*



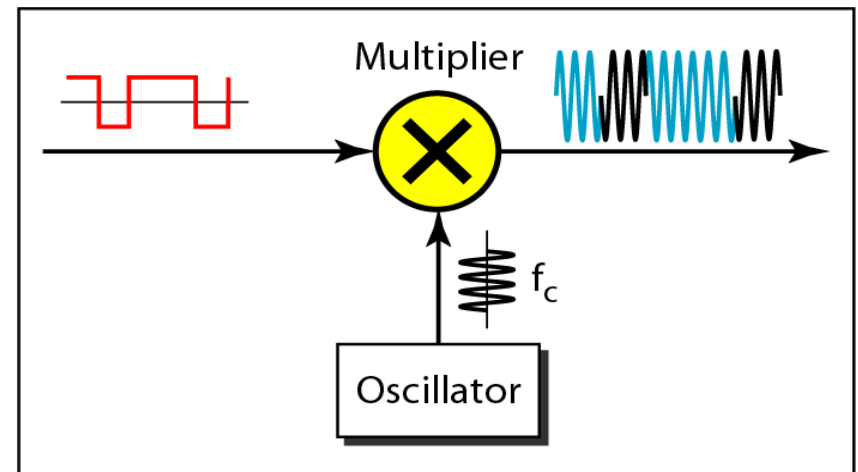
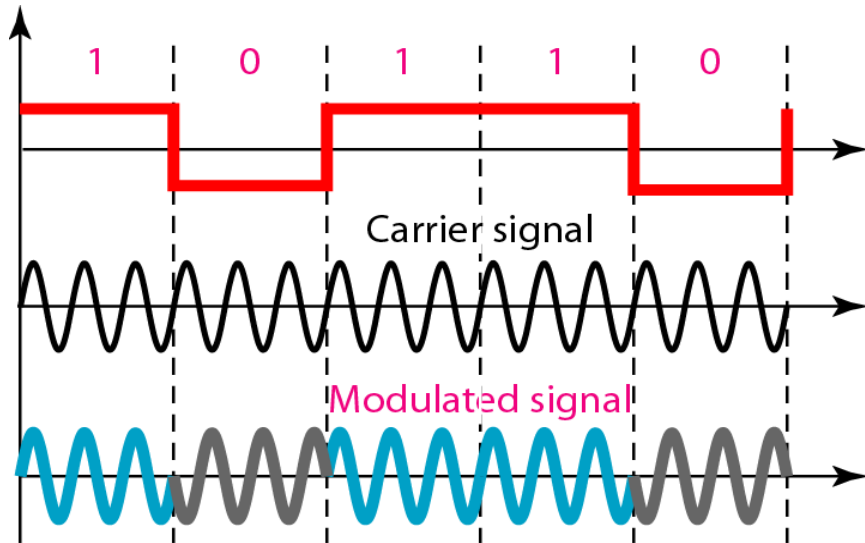
Binary phase shift keying(BPSK)

- In phase shift keying(PSK), the phase of the carrier is **varied** to represent two or more different signal elements. Both peak amplitude and frequency remain constant as the phase changes.
- Two signal elements, one with a phase of 0° , and the other with a phase of 180° .
- Binary PSK has an advantage that, it is less susceptible to noise. Noise can change the amplitude easier than it can change the phase.
- PSK is superior to FSK because we do not need two carrier signals but required sophisticated hardware to handle it.
- The bandwidth is the same as binary ASK, but less than that for BFSK.

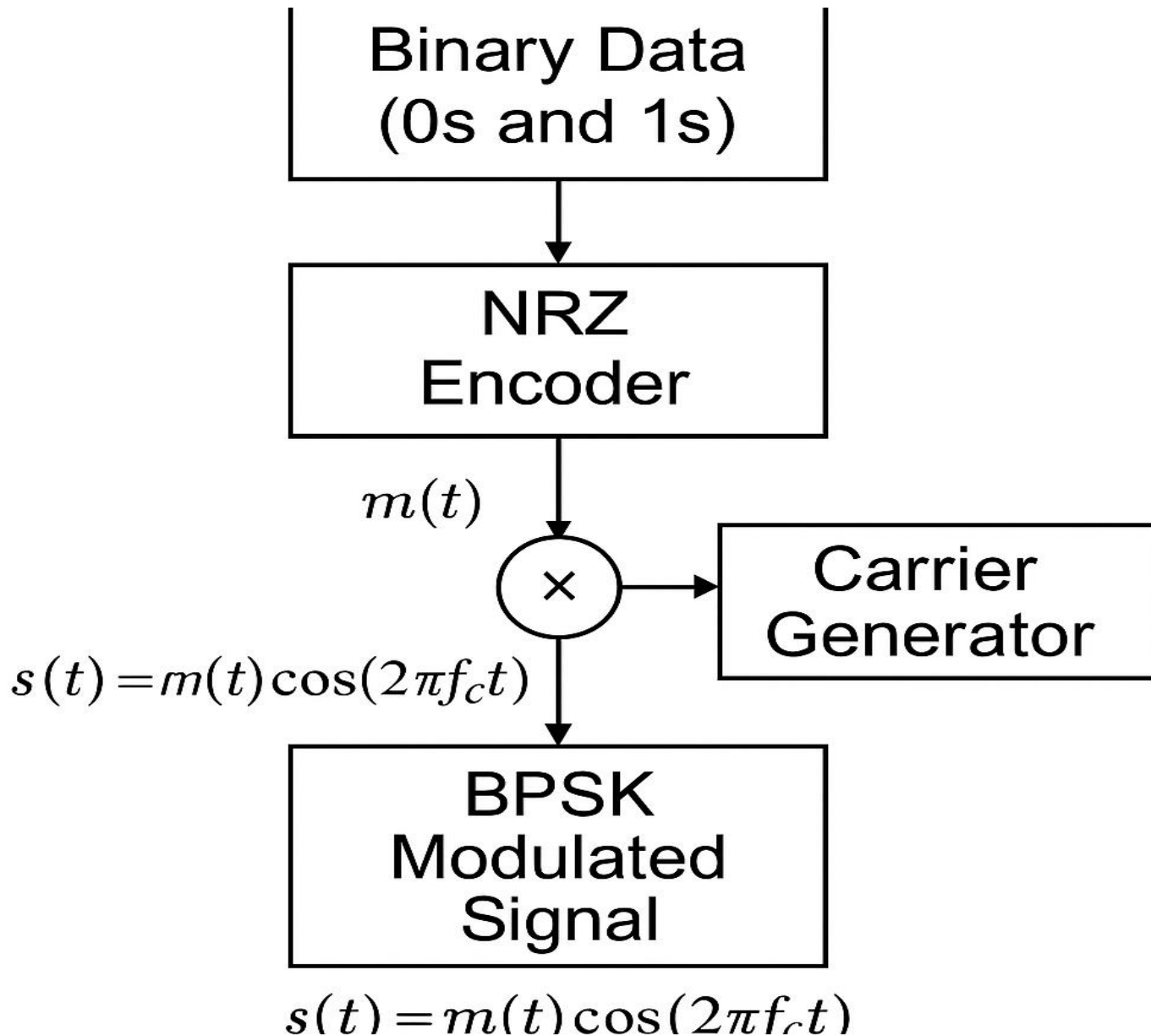


Implementation of BPSK

- The signal element with phase 180° can be seen as the complement of the signal element with phase 0°
- The polar NRZ signal is multiplied by the carrier frequency.
- The 1 bit (positive voltage) is represented by a phase starting at 0° ;
- The 0 bit (negative voltage) is represented by a phase starting at 180° .



Implementation of BPSK

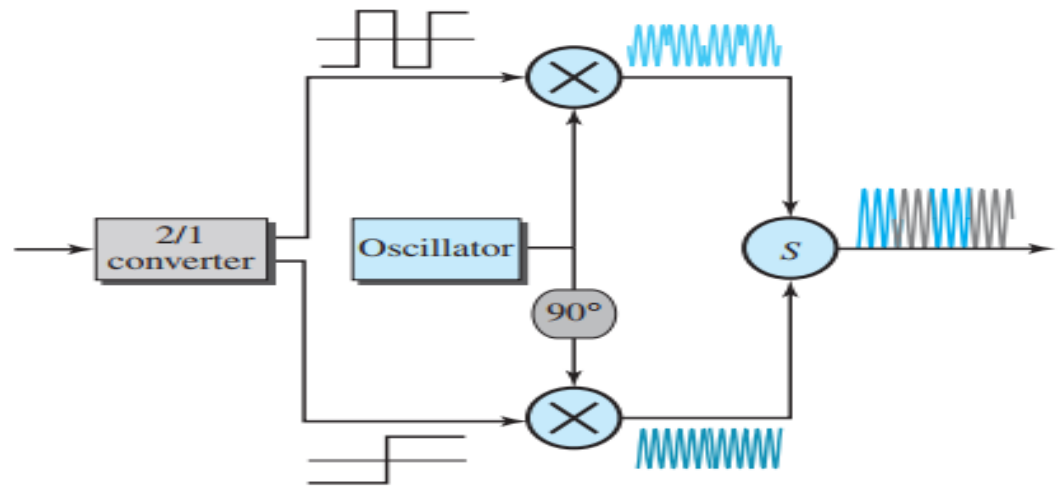
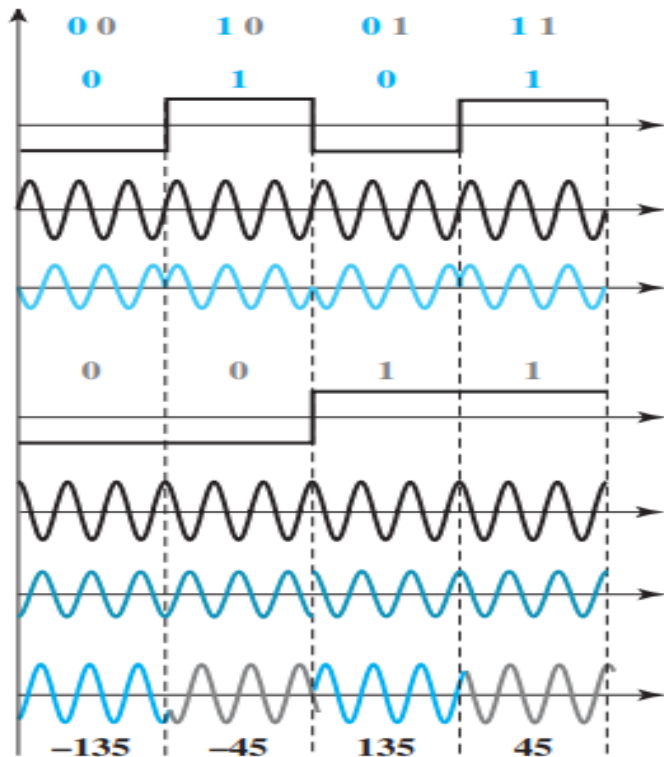


Quadrature PSK (QPSK)

- In QPSK, 2 bits sent at a time in each signal element, which decreasing the baud rate and the required bandwidth.
- The scheme is called quadrature PSK or QPSK because it uses two separate BPSK modulations; one is in-phase(Cosine), the other quadrature (out-of-phase)(Sine).
- The incoming bits are first passed through a serial-to-parallel conversion that sends one bit to one modulator and the next bit to the other modulator.
- The two composite signals created by each multiplier are sine waves with the same frequency, but different phases.
- When they are added, the result is another sine wave, with one of four possible phases: 45° , -45° , 135° , and -135°
- There are four kinds of signal elements in the output signal ($L = 4$), so we can send 2 bits per signal element ($r = 2$).

Quadrature PSK (QPSK)

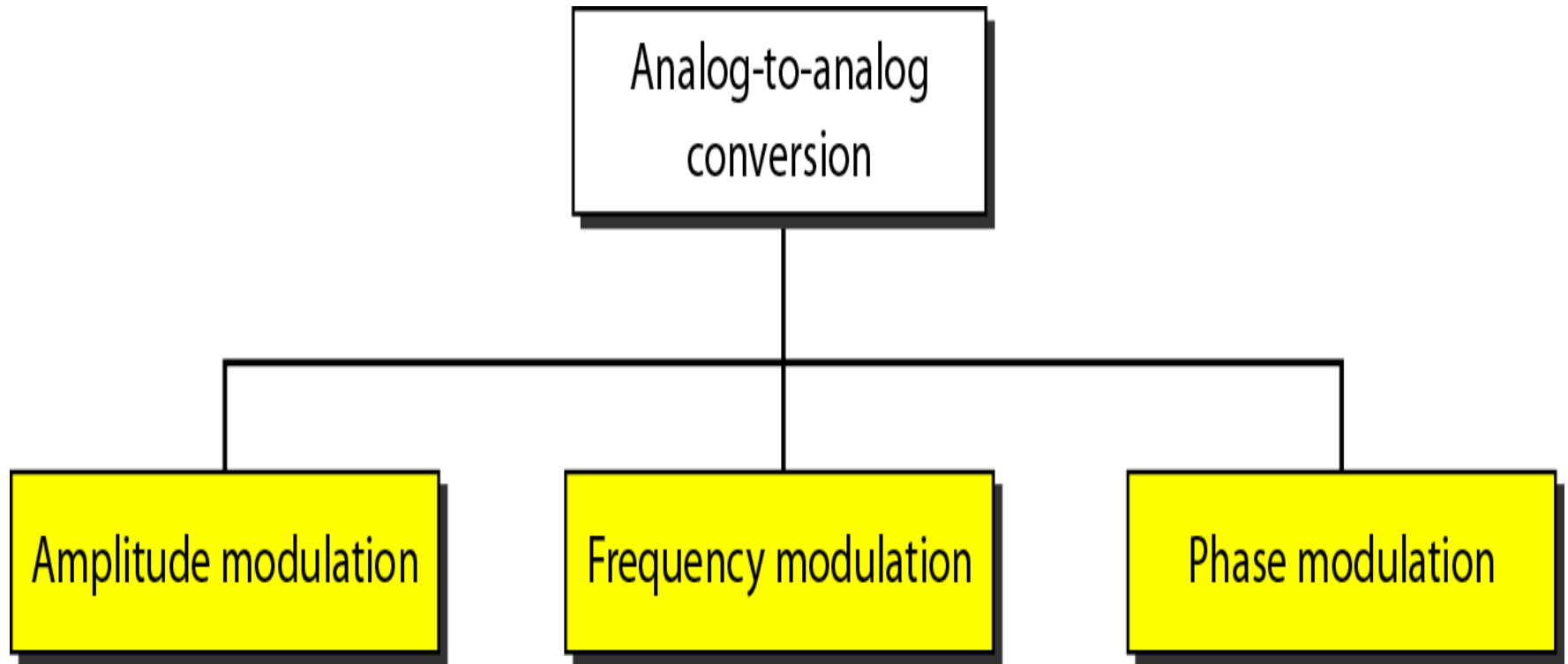
Input Bits	I	Q	Resulting Phase
00	+1	+1	+45°
01	-1	+1	+135°
11	-1	-1	-135°
10	+1	-1	-45°



5-2 ANALOG-TO-ANALOG CONVERSION

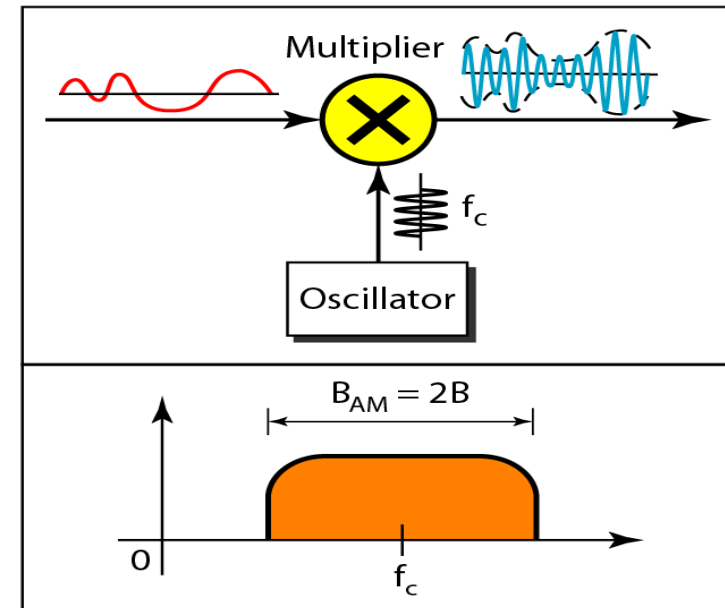
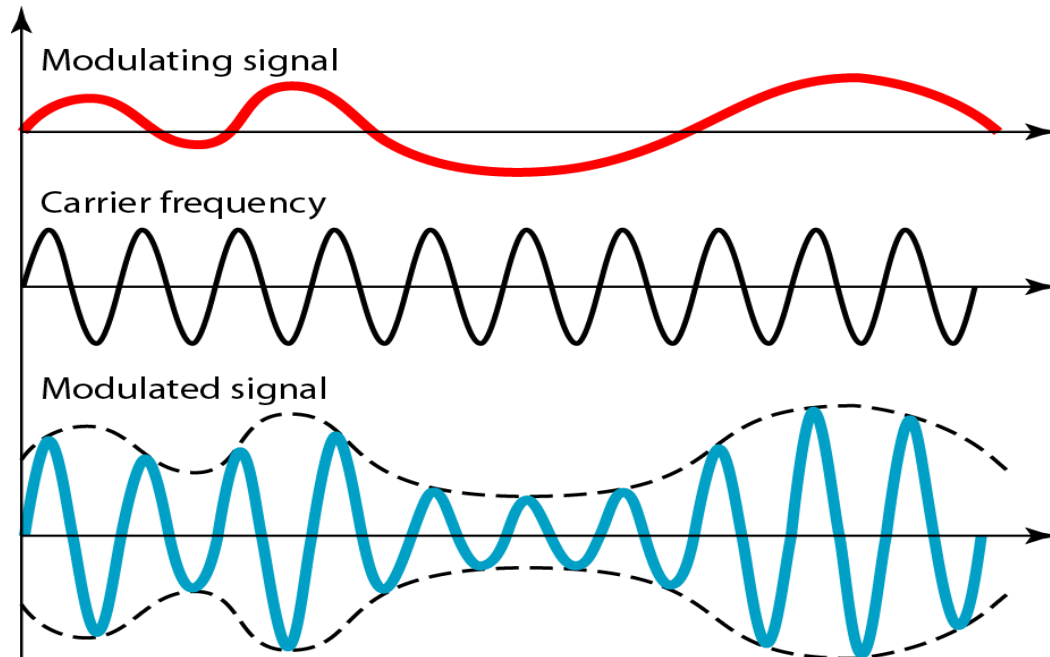
- *Analog-to-analog conversion is the representation of analog information by an analog signal. One may ask why we need to modulate an analog signal; it is already analog. Modulation is needed if the medium is bandpass in nature or if only a bandpass channel is available to us.*
- *Analog-to-analog conversion can be accomplished in three ways:*
 - **Amplitude Modulation**
 - **Frequency Modulation**
 - **Phase Modulation**

Types of analog-to-analog modulation



Amplitude modulation

- In AM transmission, the carrier signal is modulated so that its amplitude varies with the changing amplitudes of the modulating signal.
- The frequency and phase of the carrier remain the same.
- The modulation creates a bandwidth that is twice the bandwidth of the modulating signal and covers a range centered on the carrier frequency.
- The total bandwidth required for AM can be determined from the bandwidth of the audio signal: $B_{AM} = 2B$.



Amplitude modulation

- Let the message signal be:

$$m(t) = A_m \cos(2\pi f_m t)$$

- Let the carrier signal be:

$$c(t) = A_c \cos(2\pi f_c t)$$

- Then the AM signal is:

$$s(t) = [A_c + m(t)] \cos(2\pi f_c t)$$

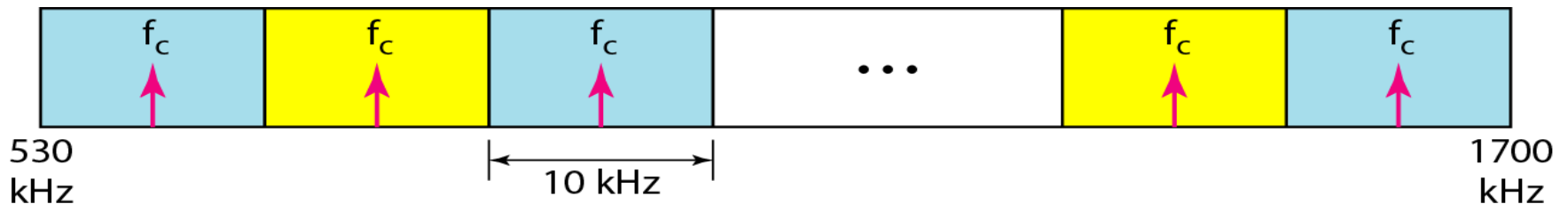
Substituting $m(t)$, $s(t) = [A_c + A_m \cos(2\pi f_m t)] \cos(2\pi f_c t)$

- The message signal $m(t)$ controls how much the frequency of the carrier deviates from its center frequency.
- AM produces three frequency components :Carrier: f_c , Upper Sideband (USB): $f_c + f_m$, and Lower Sideband (LSB): $f_c - f_m$

Condition	Value of $m(t)$	Output Amplitude
$m(t) = 0$	No message	Just carrier
$m(t) > 0$	Positive wave	Carrier amplitude increases
$m(t) < 0$	Negative wave	Carrier amplitude decreases

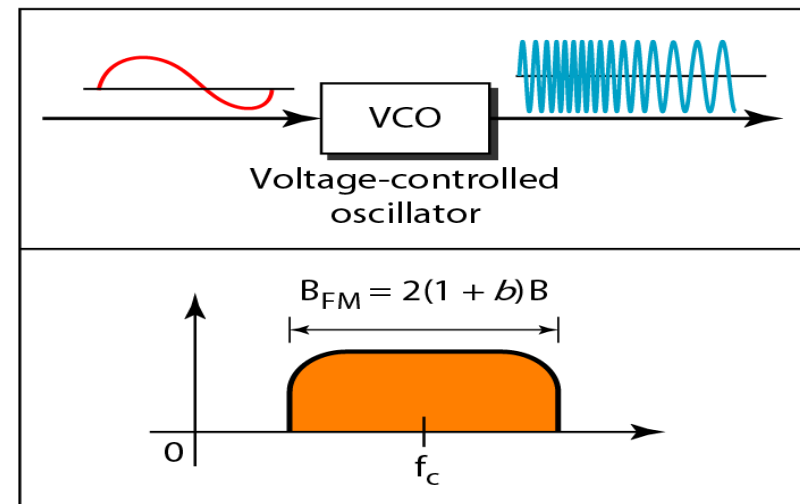
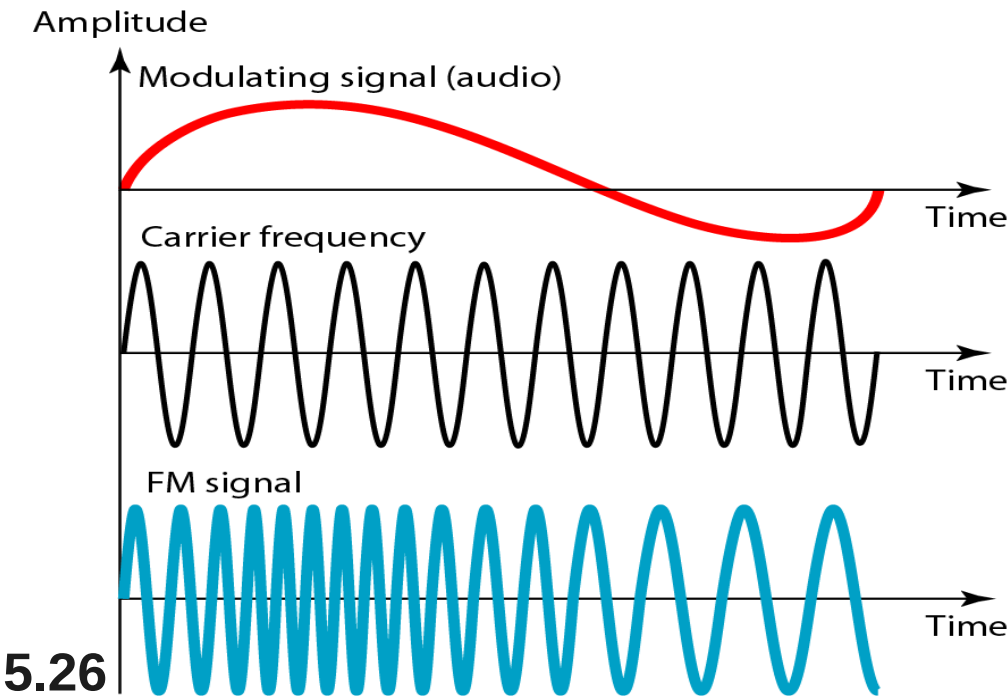
AM band allocation

- The bandwidth of an audio signal (speech and music) is usually 5 kHz. Therefore, an AM radio station needs a bandwidth of 10 kHz.
- In fact, the Federal Communications Commission (FCC) allows 10 kHz for each AM station.
- AM stations are allowed carrier frequencies anywhere between 530 and 1700 kHz (1.7 MHz).
- However, each station's carrier frequency must be separated from those on either side of it by at least 10 kHz (one AM bandwidth) to avoid interference.



Frequency modulation

- In FM, the frequency of the carrier wave is varied in proportion to the amplitude of the modulating (message) signal, while the amplitude of the carrier remains constant.
- FM is normally implemented by using a voltage-controlled oscillator as with FSK.
- The frequency of the oscillator changes according to the input voltage which is the amplitude of the modulating signal.



Frequency modulation

- The basic FM equation is

$$s(t) = A_c \cos[2\pi f_c t + 2\pi k_f \int_0^t m(\tau) d\tau]$$

Where A_c = Amplitude of carrier, f_c = Carrier frequency, k_f = Frequency sensitivity (Hz per volt), and $m(t)$ = Message signal.

- The Instantaneous Frequency of the signal is

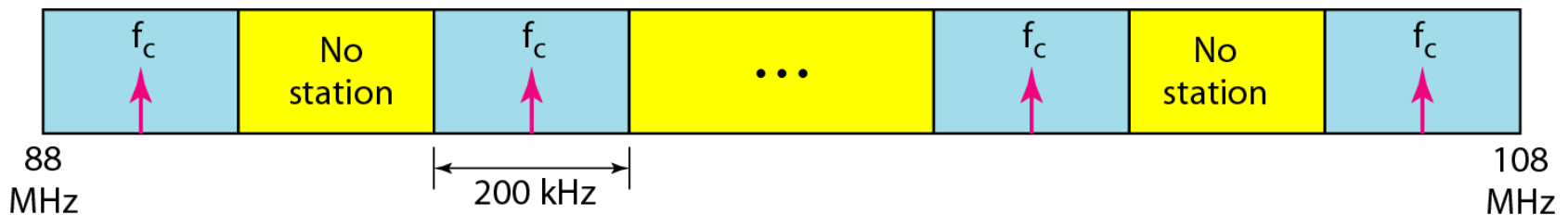
$$f_i(t) = f_c + k_f \cdot m(t)$$

- The message signal $m(t)$ controls how much the frequency of the carrier deviates from its center frequency.

$$\text{if } m(t) = \begin{cases} 0 & f_i(t) = f_c \\ > 0 & f_i(t) = f_c + k_f m(t) \\ < 0 & f_i(t) = f_c + k_f m(t) \end{cases}$$

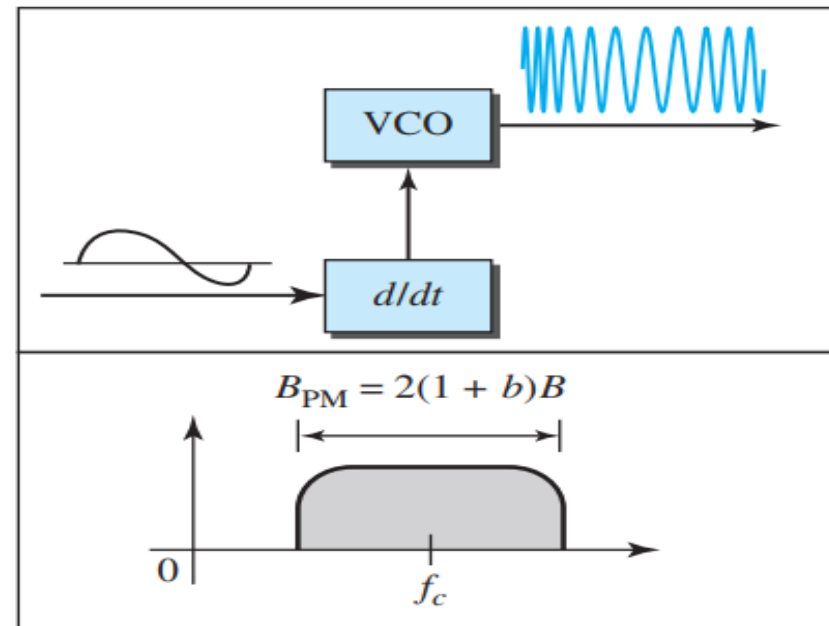
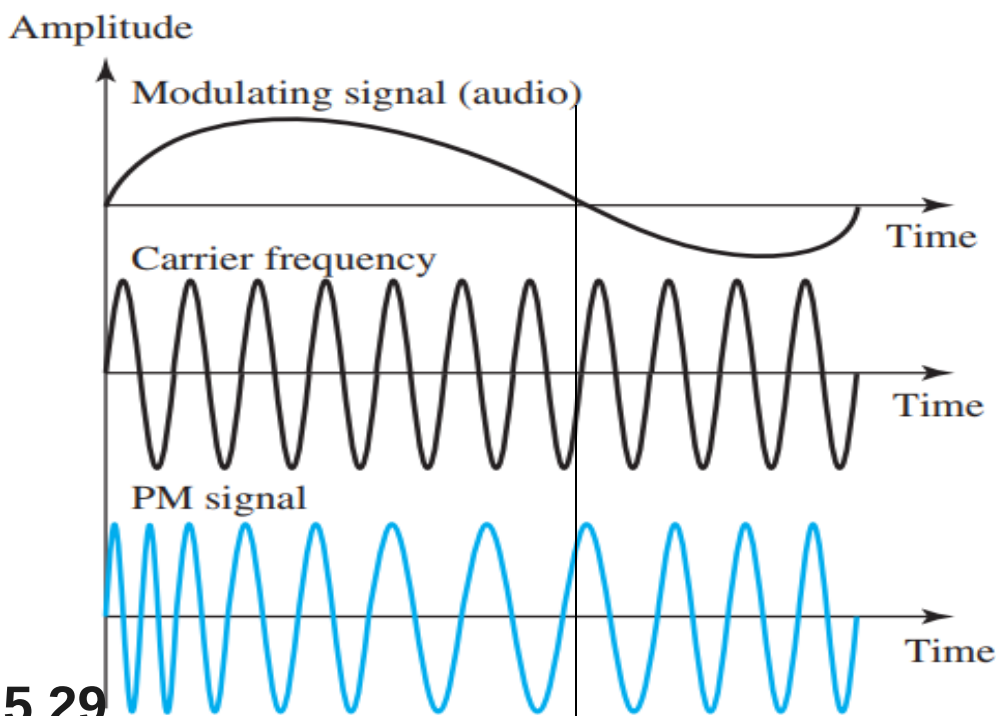
FM band allocation

- The total bandwidth required for FM can be determined from the bandwidth of the audio signal: $B_{FM} = 2(1 + \beta)B$
- The bandwidth of an audio signal (speech and music) broadcast in stereo is almost 15 kHz.
- The FCC allows 200 kHz (0.2 MHz) for each station.
- This mean $\beta = 4$ with some extra guard band.
- FM stations are allowed carrier frequencies anywhere between 88 and 108 MHz.
- Stations must be separated by at least 200 kHz to keep their bandwidths from overlapping.



Phase modulation(PM)

- The phase of the carrier signal is varied proportionally to the instantaneous amplitude of the message signal.
- The PM is the same as FM with one difference.
- In FM, the instantaneous change in the carrier frequency is proportional to the amplitude of the modulating signal, but
- In PM the instantaneous change in the carrier frequency is proportional to the derivative of the amplitude of the modulating signal



Phase modulation(PM)

- Let the message signal be $m(t)$ and the carrier signal be:

$$c(t) = A_c \cos(2\pi f_c t)$$

- Then the PM signal is:

$$s(t) = A_c \cos(2\pi f_c t + k_p m(t))$$

- Where A_c is Carrier amplitude, f_c is Carrier frequency, k_p is Phase sensitivity (radians per unit of $m(t)$), and $m(t)$ is Message signal.
- PM is normally implemented by using a voltage-controlled oscillator along with a derivative.
- The frequency of the oscillator changes according to the derivative of the input voltage, which is the amplitude of the modulating signal.
- The total bandwidth required for PM can be determined from the bandwidth and maximum amplitude of the modulating signal:

$$B_{PM} = 2(1 + \beta)B$$

- The value of β is lower in the case of PM (around 1 for narrowband and 3 for wideband).

Frequency vs Phase modulation(FM vs PM)

Feature	Frequency Modulation (FM)	Phase Modulation (PM)
Modulated Feature	Frequency of the carrier	Phase of the carrier
Instantaneous Frequency	Proportional to $m(t)$	Proportional to $d/dt [m(t)]$
Instantaneous Phase	$\phi(t) = 2\pi f_n t + k_x \int m(t) dt$	$\phi(t) = 2\pi f_n t + k_p m(t)$
Modulated Signal	$s(t) = A_c \cos[2\pi f_c t + 2\pi k_f \int_0^t m(\tau) d\tau]$	$s(t) = A_c \cos(2\pi f_c t + k_p m(t))$

Thank You